

Assignment 1: Projectile Motion

Assignment for COSC 1200 – Object-Oriented Programming 1

Prior to attempting to solve this problem, you should have viewed the Content, including assigned readings for weeks 1 through 4.

This problem is meant to be solved **without the use of generative AI**, and use of AI-generated code may result in escalation as an academic integrity alert at the professor's discretion.

Use of other reference material is encouraged. **Proof of understanding** and **proper referencing** are strongly encouraged for code copied from any other source.

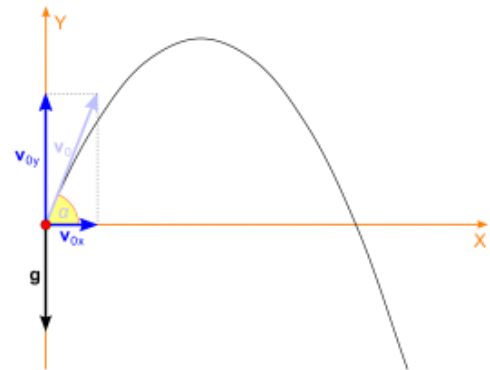
General Requirements

1. This assignment is to be completed **individually**. Group submissions will not be considered.
2. Submit your solution to the appropriate assignment folder on DC Connect **by the due date provided**.
3. Your instructor will assign a grade and post feedback on your submission to DC Connect.

Program Requirements

In this project you will create a simple Java console application that will have the user enter several values and will use these values to calculate the position of a projectile after a specific period. Our program will calculate the projectile motion by neglecting air resistance to simplify the calculations. The provided diagram represents a projectile's motion under the influence of gravity.

Formulas will be used to calculate the projectile's position at a given moment in time. Its position at any point in time (t) is calculated separately for the vertical and horizontal directions.



Horizontal – the distance travelled horizontally at time t can be calculated by:

$$d_x = (v_0 \times \cos\theta_0) t$$

Vertical – the vertical distance travelled at time t can be calculated by:

$$d_y = (v_0 \times \sin\theta_0) t - \frac{1}{2}gt^2$$

Where:

v_0 – represents the value of a projectile's initial speed in m/s at time 0

θ_0 – represents the initial angle of a projectile relative to the horizontal surface at time 0

t – represents the time that has passed in seconds

g – represents the gravitation force constant (typically -9.81 m/s^2)

Your program should prompt the user to enter the initial angle of the projectile in degrees (θ_0), followed by the initial velocity of the projectile (v_0) in m/s. Only these two inputs are required, and should be validated in such a way that the program can't crash. With valid inputs, use those inputs and the provided formulas to calculate the coordinates of the projectile. Output should show the position of the projectile at each second that passes until it hits the ground (i.e. $d_y \leq 0$), with a final output showing exactly how far it travelled (i.e. d_x).

- All user inputs may or may not contain decimal values.
- All entered values must be positive (data validation required).
- The program should provide the user very clear and concise instructions and error messages.
- All numbers in the output should be rounded to two decimal places.

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Other Requirements

- Include a comment header with your name, the date, and a short description in each human-generated code file.
- **Follow the style guide!** Use descriptive names and sensible data types for variables, constants, etc. that follow the style guide's naming conventions. Use appropriate spacing, indenting, and make sure braces ({}) are located where they should be.
- Code must be commented in line with the style guide and your professor's expectations.
- Output messages must be meaningful. Write the output as if the user has not read these instructions.
- Any assets or code obtained from, or based closely on, external sources must be cited correctly, either in line with your professor's expectations or using APA format.

Submission and Grading

To earn full marks on this assignment, you must do *all* of the following:

- Push the source code files to GitHub prior to the due date.
 - Add your professor as a collaborator on the GitHub repository that includes the source code.
- Submit the source code files to DC Connect prior to the due date.
 - Include your GitHub repository URL in the submission notes.
- Demonstrate the code file running one-on-one with your professor in the week following the due date. Your professor may allow a video or remote demonstration, or they may require you to attend class for this.

If any part of this is beyond the due date, the standard late penalty as noted in our Course Outline will apply.

The rubric for the grading of this assignment should be available on DC Connect.

Academic Integrity

By submitting this assignment to DC Connect for evaluation, you agree to the following terms:

- ☐ This work is your original work, with the majority its contents originally created by yourself.
- ☐ Any aspects of this work that were created by or based heavily on reference to another person's work, a website, a video, or a program (including generative AI systems) are cited with suitable identification to observe the original source.
- ☐ You did not transmit this file to anyone else's computer, nor receive any part of this work from anyone else's computer.

Failure to observe these points will leave you subject to Durham College's [Academic Integrity Policy](#), with a minimum penalty of a grade of 0 on the assignment and an academic integrity alert submitted to the faculty office.